Using Epigenetics to Explore Novel Treatments for Alcohol Abuse: Histone Modification in the Hippocampus and Amygdala

In 2023, 47.5% of US adults reported alcohol abuse in the past month.[1] Although many may engage in alcohol consumption without developing dependence or alcohol use disorder (AUD), alcohol use remains prevalent across communities and demographics, with 10.2% of american individuals over 12 years of age reported having AUD.[1] Alcohol abuse is associated with increased risk of accidents, greater rates of crime and violence, and increased medical and mental health costs– oftentimes due to physical and psychiatric comorbidities such as substance use, depression, anxiety, as well as circulatory system and liver complications.[2,3] Ultimately, excessive alcohol use costs U.S. society around 250 billion annually, and is one of the leading risk factors for disability and premature death— making alcohol abuse one of the most prevalent, persistent and destructive man-made contemporary public health crises.[2]

AUD is characterized by a cycle of chronic relapse where initial use is driven by its positively reinforcing effects, such as euphoria and anxiolysis, and followed by withdrawal, promoting further consumption via negative reinforcement. Stress, environmental cues, and anxiety and negative affect characteristic of alcohol withdrawal, play large roles in driving use.[2] Much research has been conducted concerning how this cycle in AUD and other substance use disorders are driven by the dysregulation of the mesolimbic dopamine system. Alcohol ingestion causes the activation of GABAa receptors in the ventral tegmental area, resulting in disinhibition of dopaminergic projections to the nucleus accumbens, mediating the reinforcing effects of alcohol.[3] This causes long lasting alterations to the system, sensitizing it to alcohol consumption. Discontinuation of use results in nervous system hyperactivity and widespread dysfunction.

Despite extensive research on the involvement of the mesolimbic dopamine system in alcoholism, as well as the environmental and behavioral correlates of AUD development and relapse, we fail to effectively treat these in the general population. Current available treatments are based on minimizing the rewarding effects of alcohol by interfering with different parts of alcohol metabolism and associated neurotransmitter systems.[3] However, the common adverse effects of these treatments limit the consistency and practicality of their use.[3] Additionally, the centrality of the mesolimbic dopamine system to all motivated and reward based behavior makes it difficult to target effectively without inducing off-target effects.[4] In consideration of the current treatment landscape and its associated nuances, attention has recently expanded to other aspects and pathways of the nervous system in pursuit of novel understanding and treatments.

In particular, the role of gene expression has received much attention in recent years—providing an interesting model through which observations of heritable susceptibility to alcoholism and the long-lasting changes induced by alcohol use on the brain may be explained. Epigenetics describes the

phenomenon of changes to gene expression that don't change the DNA sequence itself, largely done through the alteration of DNA availability. In the nucleus, gene expression is regulated by how tightly wound DNA is around histone proteins—the localized loosening of chromatin makes genes in the region accessible by transcription factors and RNA polymerase.[5] Histone acetylation and methylation mediate this, with acetylation increasing DNA availability by reducing how tightly wound DNA is around the histone, and methylation doing the opposite.[5]

Additionally, the relevant aspects of memory, negative affect/anxiety, and stress/cue induced relapse indicate that the mesolimbic dopamine system is not the only pathway contributing to alcohol use and relapse. The limbic system, specifically the amygdala and hippocampus, contribute to such behaviors through processing of emotionally salient memories and mediation of stress and anxiety through connections to the HPA axis.[6] Combining these two frontiers in neuroscience research may reveal interesting revelations concerning the pathogenesis and maintenance of AUD outside of the mesolimbic dopamine system– leading to new areas of interest for investigation and treatment. Does alcohol use drive histone modifications changes that alter limbic system structures, leading to withdrawal symptoms? What is the role of epigenetically-mediated limbic system dysfunction in AUD?

These questions can be explored through the lens of understanding the role of histone modification proteins in AUD. By understanding the pathway through which alcohol alters gene expression and how this gene expression affects limbic structures on short and long timescales, we may be able to explain the mechanisms of alcohol-induced neural dysfunction and withdrawal. Furthermore, isolating points along these pathways to target and disrupt may present promising novel treatment avenues for those with AUD.

In "A role for histone acetylation mechanisms in adolescent alcohol exposure-induced deficits in hippocampal brain-derived neurotrophic factor expression and neurogenesis markers in adulthood", Sakharkar et al. demonstrate that adolescent alcohol exposure leads to anxiogenesis and impaired hippocampal neurogenesis in adulthood, mediated by changes in histone acetylation proteins.[7] In this paper, an Adolescent Intermittent Ethanol (AIE) paradigm was used to model early alcohol use. Adolescent rats received injections of ethanol (control rats received saline) every other day for 12 days, then were allowed to mature until adulthood, when behavioral tests and molecular assays such as ELISA, immunogold, and immunohistochemical labeling, to measure protein activity and levels were performed.

AIE significantly altered the behavior of histone-modifying proteins in adulthood. Specifically, hippocampal histone deacetylase (HDAC) activity was increased more than 150%, with levels of CREB-binding protein, a histone acetyltransferase, significantly reduced in the CA1, CA2, and CA3 of the AIE hippocampus in comparison with controls. These region-specific disruptions in histone modification proteins translate into complementary reductions in levels of acetylated histone H3 lysine-9

(H3K9) at the brain-derived neurotrophic factor (BDNF) promoter, as measured by ChIP seq. Correspondingly, AIE induced decreased BDNF protein levels in the CA1 and CA3. These findings link AIE to long-lasting epigenetic changes to hippocampal histone modification proteins, translating into changes in BDNF histone association and gene expression. This is significant because the CA3 engages in reciprocal communication with the dentate gyrus, which is responsible for hippocampal neurogenesis. Additionally, BDNF is critically involved in plasticity-related neurochemical processes—implicating AIE-associated epigenetic changes to potential deficits in hippocampal development, plasticity, and function.

This is reflected in immunostaining studies of DCX and Ki-67, markers of immature and actively dividing neurons, in the AIE CA and CA3. These markers of cells engaging different stages of neurogenesis are reduced in AIE rats. However, administration of Trichostatin A (TSA), an HDAC inhibitor, rescued AIE-induced deficits in DCX, Ki-67, and acetylated histone H3K9 levels at the BDNF promoter—confirming that the described changes are a direct result of reduction in histone acetylation protein activity in AIE rats.

These findings are interesting because the CA1 directly communicates with the medial nucleus of the amygdala (MeA), and the CA3 indirectly communicates to the amygdala through this pathway. The MeA and central nucleus of the amygdala (CeA) work together to process and express fear and negative affect, as well as mediate stress and anxiety signals sent to the hypothalamus, brainstem, and dopamine systems. The researchers found that TSA also rescues AIE-induced anxiety in the light/dark box exploration and elevated plus maze tests, indicating that anxiety behavior induced by early alcohol exposure is mediated through the same histone acetylation mechanism, potentially via developmental and plasticity deficits in this hippocampal-amygdalar circuit.

Overall, this paper shows that AIE alters histone acetylation by changing activity of histone modification proteins, resulting in reduced BDNF expression and impaired hippocampal neurogenesis, coinciding with anxiogenesis. HDAC inhibition rescues both molecular and behavioral deficits (suggesting the therapeutic potential of promoting histone acetylation in AUD). These epigenetic changes may set the stage for lifelong vulnerability to anxiety and associated relapse, so isolating affected pathways in the brain will give rise to further understanding of the nature and treatment of such behavioral dysfunction. The next steps in this research are to investigate whether similar effects are seen in the amygdala and to further probe the temporal aspects of alcohol-induced epigenetic changes in these limbic structures.

You et al. examine these questions in “Reversal of deficits in dendritic spines, BDNF, and Arc expression in the amygdala during alcohol dependence by HDAC inhibitor treatment”, investigating deficits in CeA and MeA synaptic plasticity and behavior during acute alcohol withdrawal and the effect

of TSA treatment.[8] To model alcohol withdrawal, rats were fed an ethanol-enriched diet over 15-16 days before undergoing 24-hours of control diet (no ethanol). Two hours before behavioral and molecular tests, rats were injected with either TSA or a vehicle.

Gold immunolabeling revealed significant decreases in BDNF and activity-regulated cytoskeleton-associated protein (Arc) in the CeA and MeA of mice undergoing withdrawal in comparison with controls (non-ethanol, ethanol without withdrawal groups). Golgi-cox Staining reveals that these reductions in plasticity-associated proteins were accompanied by a decreased dendritic spine density within these regions, reflecting impaired synaptic plasticity and neuronal health. However, TSA administration reversed these withdrawal-induced changes, restoring BDNF and Arc levels and rescuing dendritic spine density in the CeA and MeA—demonstrating that histone acetylation mediates the synaptic and molecular deficits in these brain regions during withdrawal.

Furthermore, these changes in the amygdala were mirrored in an increase in anxiety in the withdrawal group in comparison with controls as measured by the light/dark box and elevated plus maze tests. TSA treatment mitigated these behavioral deficits. Together, these findings demonstrate that acute withdrawal reduces histone acetylation, leading to the impairment of synaptic plasticity and neuronal morphology in areas of the amygdala significant for affective and anxiogenic processing/output. These epigenetic changes are also responsible for withdrawal-induced anxiety, a key driver of relapse in AUD. Additionally, by increasing histone acetylation, HDAC inhibitors show promise in reversing the observed effects of withdrawal, potentially offering a novel therapeutic strategy to reduce withdrawal-related anxiety and, by extension, relapse rates.

These results complement those of Sakharkar et al., showing similar molecular and behavioral results on a withdrawal-based timescale. This indicates that alcohol use does not need to occur during a developmental period to induce epigenetic changes in the limbic system. Additionally, these changes specifically occur in rats undergoing withdrawal, not just during ethanol use. This suggests that ethanol use causes homeostatic changes within the limbic system that specifically reveal themselves during abstinence—contributing to the negative reinforcement cycle of withdrawal and relapse. You et al. also demonstrate that hippocampal changes in epigenetics and plasticity associated with alcohol use are paralleled in the amygdala, providing substrate for further study of how these structures interact to produce withdrawal and relapse in AUD.

Still, as these findings are born of different experimental designs, it is important to investigate epigenetic and associated changes in the amygdala in the same alcohol use paradigm as in the hippocampus. Additionally, although anxiety is a major part of alcohol withdrawal and is known to drive relapse, these behavioral results have yet to be connected to subsequent alcohol use itself. Furthermore, understanding the dynamics of how the epigenetic, plastic, and behavioral effects of discontinued alcohol

are affected during relapse will provide more insight into the role of epigenetics in AUD as well as potential therapeutic targets for further investigation.

"MicroRNA-137 Drives Epigenetic Reprogramming in the Adult Amygdala and Behavioral Changes after Adolescent Alcohol Exposure" by Kyzar et al.[9] investigates the above gaps in this research. In this study, epigenetic changes to the amygdala were measured under the same AIE paradigm as described in Sakharkar et al. However, the researchers emphasize the role of RNA in mediating observed amygdala dysregulation and its potential as a therapeutic target in AUD. Specifically, both microRNA-137 (miR-137) antagomir and Lysine-specific histone demethylase 1 (LSD1) small-interfering RNA (siRNA) were used as treatments in this study. While miR-137 is a non coding microRNA known to downregulate the expression of many genes including LSD1, LSD1 is a histone demethylase that upregulates the expression of target genes, including BDNF. Treatments were bilaterally cannulated into the CeA of rats for 2 days before testing.

AIE elevated levels of miR-137 while decreasing LSD1 mRNA levels in the adult rat amygdala, as measured by qPCR. MiR-137 antagomir administration restored LSD1 levels, revealing that AIE-induced miR-137 elevation suppresses LSD1. ChIP revealed that this dysregulation also altered BDNF expression—AIE decreased LSD1 occupancy at the BDNF promoter, while dimethylated histone H3K9 increased. These changes reduced BDNF availability and mRNA levels. Treatment with miR-137 antagomir normalized these histone modifications and restored BDNF expression, but this rescue effect was abolished by LSD1 silencing via LSD1 siRNA, affirming that miR-137 acts through LSD1 to regulate BDNF. These results confirm that the amygdala and hippocampus undergo similar epigenetic and plasticity-related changes in AIE, but what are the effects of this?

As in the previously discussed studies, AIE induces increased anxiety-like behavior in the elevated plus maze. While this was rescued by miR-137 treatment, this was likewise blocked by administration of LSD1 siRNA. These results confirm that miR-137 produces anxiogenesis by altering histone modification protein activity, perhaps exerting its effects on the limbic system through a BDNF- or plasticity-related pathway. Furthermore, the researchers connect these symptoms to subsequent alcohol use itself. Using a 2-bottle free choice preference assay where rats were allowed to freely consume either an ethanol-containing solution or water, Kyzar et al found that AIE rats had a significant preference for and consumed more ethanol than controls, but miR-137 reduced ethanol consumption to control levels for several days following administration. This dynamic regulation of AIE-induced alcohol seeking in adulthood by epigenetic changes in the amygdala through miR-137 models AUD-like behaviors and suggests that epigenetic changes induced by previous alcohol exposure potentiate use even in abstinence, mirroring the cycle of use and addiction.

Interestingly, it was also found that elevated levels of miR-137 in the AIE rat amygdala were reduced by acute ethanol administration. This suggests that alcohol preference and intake trends described here may be occurring through a withdrawal-related mechanism. These findings imply a homeostatic feedback loop where alcohol consumption temporarily alleviates dysregulation of the miR-137/LSD1/BDNF pathway in the amygdala and associated behavioral/affective effects, perpetuating the addiction cycle. Importantly, this paper demonstrates that there are two places in this circuit that can be targeted to modulate these effects—miR-137 and LSD1, suggesting that interventions targeting either of these could disrupt this cycle and potentially protect against long-lasting withdrawal symptoms and relapse vulnerability.

Kyzar et al demonstrate how miR-137-mediated histone modifications in the amygdala establish a feedback loop driving anxiety and alcohol-seeking behaviors following adolescent alcohol exposure. These insights underscore the therapeutic potential of targeting miR-137 or LSD1 to disrupt this cycle and mitigate AUD symptoms. Alongside the other papers discussed, this paints a picture of alcohol-induced deficits in hippocampal-amygdalar function and plasticity driven by changes to histone modification protein activity that appear during abstinence. Occurring both on long (developmental) and short (chronic use followed by acute withdrawal) timescales, these changes cause enduring but reversible effects on the brain and behavior—providing basis for investigation of novel therapeutic targets when attempting to reduce withdrawal and relapse in AUD.

The work of Sakharkar et al., You et al., and Kyzar et al. collectively paints a compelling narrative: Alcohol use drives enduring changes in histone modification protein activity that leads to deficits in histone acetylation in the amygdala and hippocampus. These epigenetic modifications alter protein expression and plasticity in the limbic system while also contributing to behaviors related to AUD development and relapse such as anxiety and alcohol consumption/preference mediated by alcohol withdrawal/abstinence. Together, these findings provide significant insights into the epigenetic mechanisms underlying AUD while highlighting alternative neuroanatomical pathways that contribute to AUD, offering promising directions for future research and potential treatments.

Different aspects of epigenetic histone acetylation and miR-mediated regulation converge on common pathways, particularly affecting BDNF and other markers of neuroplasticity, such as Arc, Ki-67, and DCX. These changes appear to alter the structural and functional integrity of the hippocampus and amygdala, contributing to behavioral symptoms like anxiety and reinforcing further alcohol use through negative reinforcement. Interestingly, mutations in the BDNF and other plasticity-related genes have been identified as risk factors for AUD—conferring early onset of relapse in mouse studies that is rescued by overexpression of wild-type BDNF or activation of TrkB, the BDNF receptor.[3] These findings strengthen

the argument that the epigenetic dysregulation of neuroplasticity-related proteins plays a major role in AUD.

Epigenetic interventions show promise for disrupting this cycle. HDAC inhibitors, as demonstrated in Sakharkar et al. and You et al., reverse deficits in histone acetylation and rescue associated molecular and behavioral symptoms in animal models. Similarly, Kyzar et al. highlight the potential of targeting miR-137 or its downstream effectors, such as LSD1, to restore proper epigenetic regulation, mitigate anxiety, and reduce alcohol preference. Together, these studies suggest that manipulating nodes along these epigenetic pathways can address the molecular underpinnings of AUD, paving the way for novel therapeutic strategies.

However, several questions remain. While these studies link histone modification and microRNA dysregulation to neural and behavioral deficits, additional research is needed to identify the precise genes affected by these epigenetic changes and how they directly impact neuronal circuits. While BDNF emerges as a key player, the exact dynamics of its dysfunction in AUD are not explored here. Other genes involved in synaptic plasticity and stress response might also be critical targets. Similarly, the temporal dynamics of epigenetic changes in relation to withdrawal and relapse warrant further investigation. How do these changes evolve over time? Are they permanent, or can they be reversed under specific conditions? Furthermore, the behavioral models used in these studies fail to accurately model human alcohol use, as the initial ethanol exposure in these studies is forced rather than being based on free consumption. Other aspects of AUD and relapse, such as the effect of stress, alcohol-associated cues, and other environmental factors also remain unaddressed in these studies.

Future experiments should expand upon the paradigms used in these studies to address these gaps. Longitudinal studies investigating the persistence of epigenetic changes across different periods of abstinence and relapse will be crucial for understanding the temporal relationship between alcohol use, withdrawal, and plasticity-related deficits. Additionally, transcriptomic approaches with proteomic and behavioral assays could identify new epigenetically regulated targets within the hippocampus and amygdala.

In conclusion, the discussed studies underscore the centrality of hippocampal and amygdalar epigenetics in AUD, perpetuating the addiction cycle. By identifying and targeting these pathways, we can advance our understanding of AUD and develop novel therapeutic approaches to mitigate withdrawal symptoms, reduce relapse, and improve treatment outcomes.

Citations

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